

Design and Simulation of a Telescopic CMOS Operational Amplifier in 180 nm Technology

Parthoy Narzary T25133, Rohak Gupta B23500, Himanshu Kumar S25027

Group 14

School of Computing and Electrical Engineering, IIT Mandi

Abstract—This work presents the design and Cadence Virtuoso implementation of a single-stage telescopic operational amplifier in the 180 nm SCL CMOS process. The amplifier is intended to drive a 1 pF load capacitor with a target DC gain of 40 dB and a unity gain bandwidth (UGBW) of 1 GHz at 1.8 V supply. Hand calculations based on the gain–bandwidth relation $GBW \approx g_{m1}/(2\pi C_L)$ and assumed internal node voltages are used to estimate the required bias currents and transistor aspect ratios. The final design operates with a tail current of approximately 700 μ A and achieves a simulated DC gain of 56.43 dB, UGBW of 736.54 MHz, phase margin of 82.7°, and slew rate of 331.93 V μ s^{−1}. Additional simulations report an output-referred noise density of 3.335 nV/ $\sqrt{\text{Hz}}$, PSRR of about 34.9 dB, CMRR of 11.3429 dB, an input common-mode range of 254.3 mV, and an output common-mode range of 2.02 mV. These results illustrate both the strengths (high gain and speed) and limitations (restricted signal swing and modest CMRR/PSRR) of telescopic topologies in a practical design flow.

Index Terms—telescopic op-amp, CMOS analog design, gain-bandwidth product, slew rate, PSRR, CMRR

I. INTRODUCTION

Telescopic CMOS operational amplifiers are widely used as high-speed, low-noise gain stages in data converters and switched-capacitor circuits [1], [2]. Compared to simple differential pairs, telescopic cascode structures offer significantly higher intrinsic gain due to the product of multiple transconductances and output resistances, while consuming only a single bias current. However, the stacking of transistors severely limits the available signal swing and the input/output common-mode ranges. Designing such an amplifier in a modern CMOS technology therefore requires a careful allocation of headroom, overdrive voltages, and bias currents.

This project focuses on the design of a single-stage telescopic op-amp in a 180 nm SCL CMOS process using Cadence Virtuoso and Spectre. The primary specification is to realize at least 40 dB DC gain and approximately 1 GHz unity-gain bandwidth while driving a 1 pF capacitive load. In addition to the main frequency-domain metrics, practical figures such as slew rate, PSRR, CMRR, noise, and common-mode ranges are evaluated.

From a practical viewpoint, the design flow combines textbook device-level theory with Cadence-specific steps: (1) starting from pen-and-paper sizing using simple equations, (2) verifying DC operating points and overdrive voltages, (3) measuring small-signal gain and GBW using AC analysis in a closed-loop configuration, (4) extracting large-signal slewing

using transient simulations, and (5) iterating device sizes and bias voltages through a separate biasing network.

Throughout this work, standard analog MOS models from the 180 nm SCL kit are used. Unless mentioned otherwise, simulations assume a supply voltage of 1.8 V and a nominal temperature of 27 °C.

II. TELESCOPIC OP-AMP ARCHITECTURE

A. Core Amplifier Topology

The chosen topology is a classical NMOS-input telescopic op-amp. A conceptual schematic of the core is shown in Fig. 1. The main elements are:

- A differential NMOS input pair M_1 – M_2 .
- NMOS cascode devices M_3 – M_4 to boost output resistance.
- PMOS current-mirror loads M_5 – M_8 forming the high-resistance upper stack.
- A tail NMOS current source M_{tail} that defines the differential pair bias current.

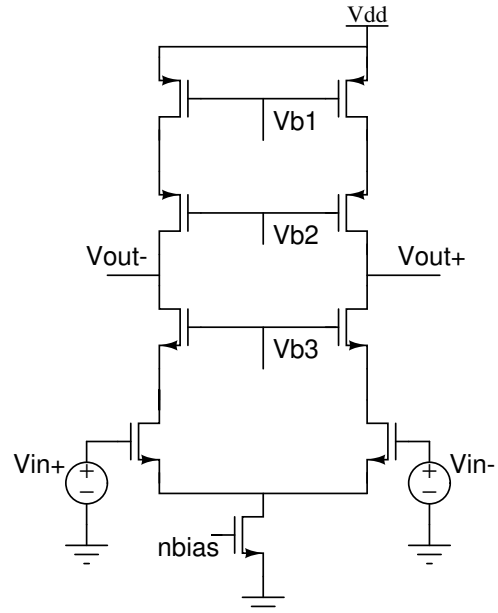


Fig. 1: Schematic of the designed telescopic operational amplifier (core stage).

The DC gain of the amplifier can be approximated as

$$A_v \approx g_{m1} (r_{o,\text{upper}} \parallel r_{o,\text{lower}}), \quad (1)$$

where g_{m1} is the transconductance of the input pair and $r_{o,upper}$, $r_{o,lower}$ are the effective output resistances of the upper PMOS and lower NMOS stacks, respectively. The telescopic structure effectively multiplies the intrinsic transistor resistances, leading to gains well above 40 dB even in 180 nm.

TABLE I: Transistor Dimensions Used in the Telescopic Op-Amp

Device	W (μm)	L (μm)
Input NMOS (M1, M2)	42.94	0.18
NMOS Cascode (M3, M4)	36.77	0.36
Tail NMOS (Mtail)	31.82	0.18
PMOS Load (M5, M6)	29.29	0.18
PMOS Cascode (M7, M8)	76.26	0.36

B. Biasing Network

To generate the required gate bias voltages for telescopic devices, a dedicated biasing network is used (Fig. 2). The network creates three main bias voltages:

- V_{B3} for the top PMOS cascodes,
- V_{B2} for the PMOS mirror devices,
- V_{B1} for the NMOS cascodes and tail.

The final simulated bias voltages obtained from the biasing network were:

$$V_{B1} = 1.135 \text{ V}, \quad V_{B2} = 657.58 \text{ mV}, \quad V_{B3} = 492.7 \text{ mV},$$

$$V_{nbias} = 650 \text{ mV}, \quad V_{in,CM} = 725 \text{ mV}.$$

These voltages were used throughout all AC, noise, PSRR, and transient simulations.

The bias generator is itself biased from a reference current, chosen to be comparable to the tail current of the main amplifier so that the overdrive voltages match.

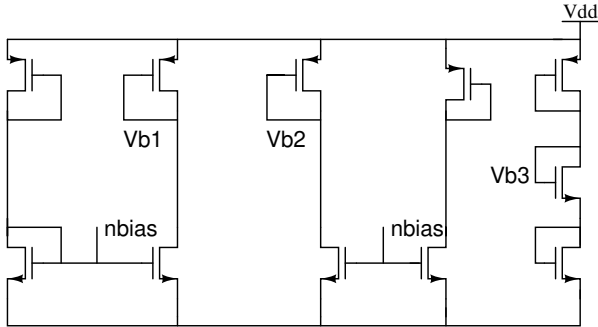


Fig. 2: Biasing network generating V_{B1} , V_{B2} , and V_{B3} for the telescopic core.

During early simulations the telescopic core was driven directly by ideal DC voltage sources at the bias nodes to validate the hand calculations. After introducing the transistor-level biasing network, the effective node voltages shifted slightly and the tail current settled around $700 \mu\text{A}$, instead of the initial design target of $900 \mu\text{A}$. The final performance numbers presented later correspond to this fully biased configuration.

III. DESIGN METHODOLOGY

A. Initial Specifications

The main design specifications are summarized in Table II. Only gain, GBW and load are hard targets; the other metrics are left to be observed and optimized as the design evolves.

TABLE II: Initial Design Specifications

Quantity	Symbol	Target
Supply voltage	V_{DD}	1.8 V
Load capacitance	C_L	1 pF
DC gain	A_v	$\geq 40 \text{ dB}$
Unity-gain bandwidth	GBW	$\approx 1 \text{ GHz}$
Topology	–	Single-stage telescopic
Technology	–	180 nm SCL CMOS

B. Hand Calculations

1) Transconductance from GBW

For a single-pole amplifier driving a capacitive load,

$$GBW \approx \frac{g_{m1}}{2\pi C_L}. \quad (2)$$

For $GBW_{\text{target}} = 1 \text{ GHz}$ and $C_L = 1 \text{ pF}$,

$$g_{m1,\text{target}} \approx 2\pi \cdot 1\text{pF} \cdot 1\text{GHz} \approx 6.28 \text{ mS}. \quad (3)$$

An NMOS overdrive $V_{ov,n}$ in the range 200 mV to 250 mV was considered reasonable for speed and noise. Using the relation $g_m = 2I_D/V_{ov}$, the differential pair current becomes

$$I_{D,\text{pair}} \approx \frac{g_{m1} V_{ov,n}}{2}, \quad (4)$$

yielding an initial estimate close to $900 \mu\text{A}$.

2) Headroom and Node Voltages

Because telescopic structures stack multiple devices between V_{DD} and ground, approximate node voltages were fixed in advance to guarantee saturation:

$$\begin{aligned} V_{DD} &= 1.8 \text{ V}, \\ V_{N1} &\approx 1.35 \text{ V}, \\ V_{N2} &\approx 0.9 \text{ V}, \\ V_{N3} &\approx 0.6 \text{ V}, \\ V_{N4} &\approx 0.3 \text{ V}. \end{aligned}$$

Here V_{N1} to V_{N4} represent descending internal nodes of the cascode stack. Assuming $V_{th} \approx 450 \text{ mV}$, the gate bias voltages are approximated as

$$V_{G,n} \approx V_S + V_{th,n} + V_{ov,n}, \quad (5)$$

$$V_{G,p} \approx V_S - V_{th,p} - V_{ov,p}. \quad (6)$$

Several values of $V_{ov,p}$ around 0.4 V to 0.45 V were tried for the PMOS devices so that they could sustain the required current in saturation.

These equations directly produced first-pass bias voltages V_{B1} , V_{B2} and V_{B3} and rough aspect ratios for all devices. The PMOS overdrives were intentionally kept larger than the NMOS overdrives to simplify matching of currents and to provide some margin for channel length modulation.

3) Gain Estimation

Assuming a simple $g_m r_o$ model for each device and a modest Early voltage V_A , the small-signal gain of the telescopic op-amp can be approximated as

$$A_v \approx g_{m1}r_{o1} \cdot g_{m3}r_{o3} \parallel g_{m5}r_{o5} \cdot g_{m7}r_{o7}. \quad (7)$$

Using hand estimates of V_A and the calculated currents, the expected gain comfortably exceeded the 40 dB target, giving confidence that the topology was suitable.

C. Simulation Procedure

All circuits were implemented in Cadence Virtuoso with Spectre as the simulator. The main steps were:

- 1) **DC operating point check:** Verify that each device stays in saturation with the assumed node voltages; adjust W/L and bias voltages as needed.
- 2) **AC analysis for differential gain and UGBW:** Connect the op-amp in a unity-gain feedback configuration and run small-signal AC analysis to obtain the Bode plot (Fig. 4). The unity-gain frequency and phase margin are read from this plot.
- 3) **Transient analysis for slew rate:** Apply a large input step in the closed-loop configuration and measure the maximum slope of the output (Fig. 11).
- 4) **Noise analysis:** Use Spectre's noise analysis to compute output-referred spectral density across frequency (Fig. 9).
- 5) **PSRR, CMRR and common-mode range:** Simulate common-mode gain, supply perturbations, and DC sweeps of the input/output common-mode voltages (Figs. 6–8).

IV. RESULTS AND DISCUSSION

This section presents the simulated differential gain, phase response, and frequency characteristics of the telescopic operational amplifier.

A. Differential Gain and Frequency Response

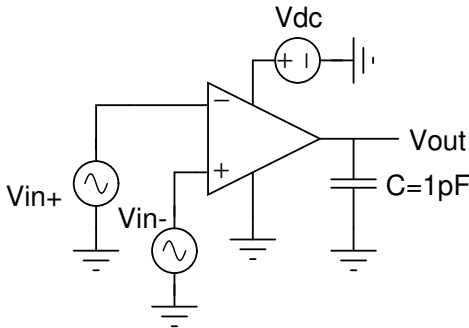


Fig. 3: Testbench schematic used for AC differential gain and phase simulation in unity-gain feedback.

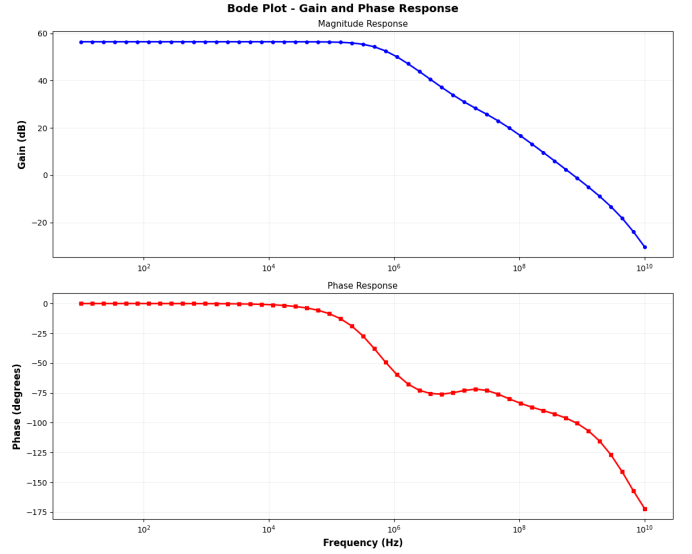


Fig. 4: Simulated differential gain and phase versus frequency for the telescopic op-amp in unity-gain feedback.

Figure 4 shows the simulated magnitude and phase of the closed-loop response. From the low-frequency region, the extrapolated DC open-loop gain is approximately 56.43 dB, significantly higher than the 40 dB requirement. The magnitude crosses 0 dB around 736.54 MHz, which is taken as the UGBW. The phase at unity gain is about 82.7°, providing a comfortable stability margin for capacitive loads and mild variations in process or bias.

The slightly lower-than-target GBW (relative to 1 GHz) is attributed mainly to the reduced actual tail current (700 μ A instead of 900 μ A) and increased parasitic capacitances from the finalized device sizes and biasing network.

B. Common-Mode Rejection Ratio

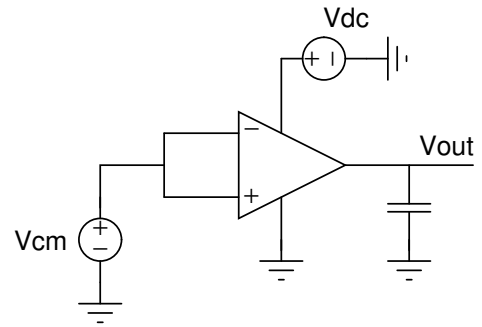


Fig. 5: Testbench schematic for extracting common-mode gain and CMRR. A common-mode AC source is applied while differential feedback is maintained.

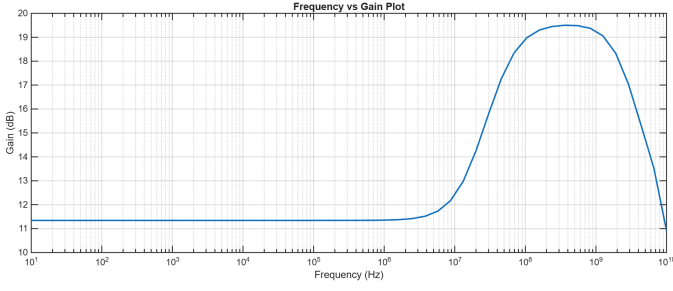


Fig. 6: Simulated common-mode gain and derived CMRR across frequency.

Figure 6 depicts the common-mode gain response. The resulting CMRR at low frequency is about 11.3429 dB, which is relatively modest for a precision operational amplifier. This limited CMRR is mainly due to the single current source at the tail and the unavoidable mismatch between the upper PMOS current mirrors and lower NMOS stacks in a strictly telescopic structure. Improved CMRR would require either more accurate current sources, additional common-mode feedback, or a fully differential output stage with dedicated common-mode control.

C. Power-Supply Rejection Ratio

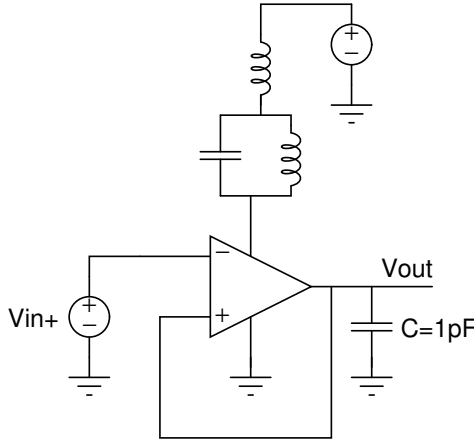


Fig. 7: PSRR testbench schematic, where an AC signal is injected at the supply node V_{DD} while the amplifier output is monitored.

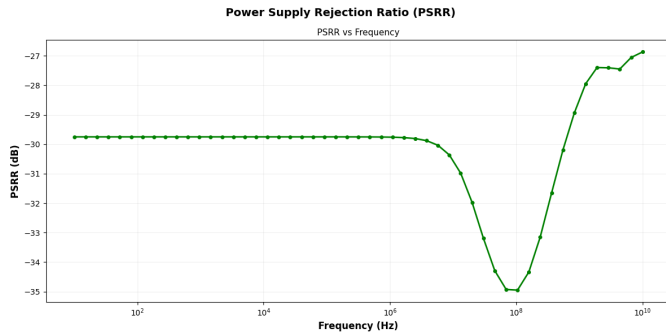


Fig. 8: Simulated power-supply rejection ratio (PSRR) versus frequency.

The PSRR plot in Fig. 8 shows how disturbances on the supply couple to the output. At low frequencies, the PSRR magnitude is approximately -34.9 dB. As frequency increases, the supply rejection degrades due to parasitic coupling through the gate-source capacitances and the finite impedance of the biasing network. For many on-chip applications with relatively clean, regulated supplies this level of PSRR is acceptable, although it is lower than what can be achieved with multi-stage or gain-boosted architectures.

D. Noise Performance

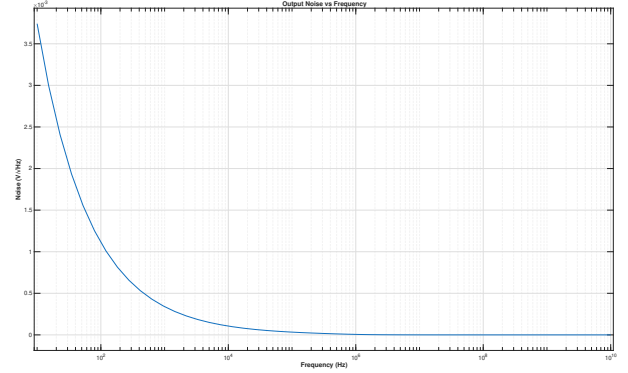


Fig. 9: Output-referred noise spectral density of the telescopic op-amp.

Figure 9 illustrates the simulated output noise spectral density. At midband frequencies beyond the $1/f$ corner, the noise flattens to about $3.335 \text{ nV}/\sqrt{\text{Hz}}$. The low-frequency noise is dominated by the input NMOS pair and tail device, as expected. For applications such as switched-capacitor circuits where the signal band lies well above the $1/f$ corner, the wideband noise figure is more relevant and the obtained value is competitive for a single-stage amplifier in 180 nm technology.

E. Slew Rate

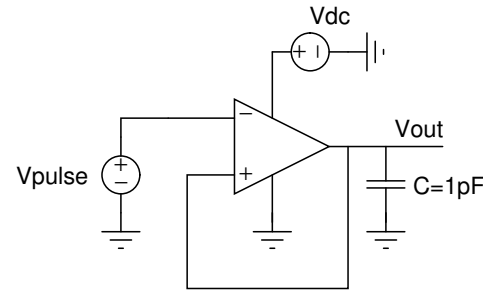


Fig. 10: Schematic of the testbench used for slew-rate simulation. The amplifier is connected in unity-gain feedback and driven with a large input step.

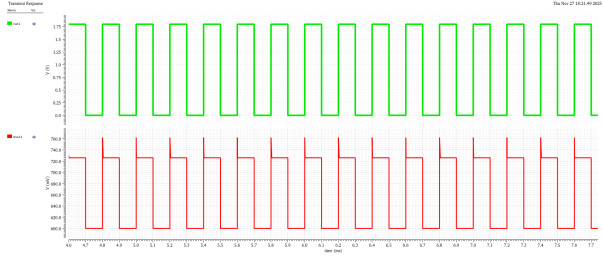


Fig. 11: Transient response to a large input step used for slew-rate extraction.

The large-signal transient in Fig. 11 demonstrates the output response to a substantial input step in a unity-gain configuration. By measuring the maximum slope of the rising edge, the slew rate is found to be approximately $331.93 \text{ V } \mu\text{s}^{-1}$. This high value is consistent with the relatively large tail current and small load capacitance, since

$$SR \approx \frac{I_{\text{available}}}{C_L}. \quad (8)$$

The result confirms that, once the amplifier saturates into current-limited operation, it can charge and discharge the 1 pF load very rapidly.

F. Common-Mode Ranges

Input and output common-mode ranges were extracted from DC sweeps. The input common-mode range (ICMR) is approximately 254.3 mV . The output common-mode range (OCMR) is extremely narrow, around 2.02 mV , which is expected from a telescopic stack where several overdrive voltages must be maintained simultaneously. These results clearly illustrate one of the classic limitations of telescopic op-amps: excellent gain and speed at the cost of limited swing. In practice, such an amplifier is typically embedded in closed-loop configurations where the common-mode is tightly controlled and the required swing is modest.

V. LESSONS LEARNED AND DESIGN ISSUES

Several practical issues arose during the design:

- **Bias network interaction:** The first set of simulations used ideal DC sources for bias voltages, which gave a tail current close to the hand-calculated $900 \mu\text{A}$. When the transistor-based biasing network was connected, the bias voltages shifted and the operating current settled around $700 \mu\text{A}$. This reduced the actual g_m and hence the UGBW, demonstrating the importance of including realistic bias circuitry early in the design.
- **Headroom trade-offs:** Trying to push for larger swing by reducing some overdrive voltages led to devices falling out of saturation in DC operating point checks. Recovering saturation required either increasing device lengths or decreasing the tail current, both of which impacted GBW and noise.
- **CMRR/PSRR limitations:** Attempts to improve CMRR and PSRR by re-sizing the bias devices often conflicted

with the gain and headroom requirements. This highlighted the intrinsic trade-offs of the simple telescopic topology and motivated thoughts about gain-boosting and common-mode feedback for future work.

VI. APPLICATIONS AND FUTURE IMPROVEMENTS

Thanks to its relatively high gain, sub-GHz unity-gain bandwidth, and low output noise, the designed telescopic amplifier is suitable for use as:

- The first gain stage in moderate-speed pipeline or SAR ADCs where large swings are not required.
- A preamplifier for low-capacitance sensor interfaces in which the output common-mode is set by on-chip biasing.
- A building block inside switched-capacitor filters or integrators with tightly controlled common-mode levels.

Future improvements could include:

- Adding a dedicated common-mode feedback loop to stabilize the output common-mode voltage and extend the usable signal range.
- Employing gain-boosting amplifiers on the cascode devices to increase both gain and PSRR without significantly changing the basic topology.
- Performing full corner, Monte-Carlo, and layout-parasitic-extracted simulations to evaluate robustness under process, voltage, and temperature variations.
- Exploring alternative architectures such as folded cascode or two-stage op-amps for applications that require wider input/output ranges at similar bandwidths.

SUMMARY OF MEASURED PERFORMANCE

TABLE III: Final Measured Performance versus Targets

Metric	Symbol	Target	Measured
Supply voltage	V_{DD}	1.8 V	1.8 V
Load capacitance	C_L	1 pF	1 pF
Tail current	I_{tail}	$\approx 0.9 \text{ mA}$	0.7 mA
DC gain	A_v	$\geq 40 \text{ dB}$	56.43 dB
Unity-gain BW	GBW	$\approx 1 \text{ GHz}$	736.54 MHz
Phase margin	PM	$> 60^\circ$	82.67°
Slew rate	SR	—	$331.93 \text{ V } \mu\text{s}^{-1}$
Output noise	$e_{n,\text{out}}$	—	$3.335 \text{ nV}/\sqrt{\text{Hz}}$
PSRR	—	—	-34.93 dB
CMRR	—	—	11.3429 dB
ICMR	—	—	254.32 mV
OCMR	—	—	2.02 mV

VII. CONCLUSION

A single-stage telescopic CMOS operational amplifier was designed and simulated in a 180 nm SCL technology using Cadence Virtuoso. Starting from textbook expressions and assumed node voltages, bias currents and transistor sizes were estimated, and a complete bias network was implemented to generate the required gate voltages. After accounting for the interaction between the core and the bias network, the final design operated with a tail current of about $700 \mu\text{A}$.

The amplifier achieved a simulated DC gain of 56.43 dB , UGBW of 736.54 MHz , and phase margin of 82.7° while

driving a 1 pF load. Large-signal simulations indicated a slew rate of $331.93 \text{ V } \mu\text{s}^{-1}$, and noise analysis reported an output-referred density of $3.335 \text{ nV}/\sqrt{\text{Hz}}$. Measured PSRR and CMRR were around 34.9 dB and 11.3429 dB, respectively, and the common-mode ranges were found to be quite limited, consistent with the telescopic topology.

Overall, the project demonstrated how classical design equations from Razavi and Allen–Holberg could be combined with modern CAD tools to realize a high-gain, moderately high-speed amplifier while revealing the practical trade-offs between gain, bandwidth, headroom, and rejection ratios.

REFERENCES

- [1] B. Razavi, *Design of Analog CMOS Integrated Circuits*. McGraw–Hill, 2001.
- [2] P. E. Allen and D. R. Holberg, *CMOS Analog Circuit Design*, 2nd ed. Oxford Univ. Press, 2002.
- [3] Cadence Design Systems, *Virtuoso Design Environment User Guide*, 2025.